AI, Your Brain, and the Tripos

Guidance on Effective and Responsible Use

Will Handley, on behalf of the Institute of Astronomy

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TL;DR: use this time to train your brain

The next one or two years are a golden opportunity

Make the most of it

You have time, difficult problems, other students and supervisors to help you develop your own understanding. Don't spend that opportunity prompting an AI to do the thinking you came here to practise.

- Human brains will still be needed, particularly those able to recognise and creatively develop the opportunities AI brings. AI can help build that understanding, but it can also compromise it.
- Ideally, supervision work should be entirely your own unaided work. Where it is not, make clear to your supervisor what assistance you used.
- AI is optional. The department supports your decision if you choose not to use it.

An apparent paradox: AI research and unaided learning

The IoA and wider Cambridge astronomy are leaders in using AI for research





Comment

Large language models are not the problem

Hiranya V. Peiris

Check for updates

There is currently a frenetic and occasionally ill-informed debate around text-to-community about large language models (LLMs). I have been following my settlement on, not least because I work in the space of AI and cosmology. Here, I explain why I think that LLMs are powerful models for galaxy population and simulation emulation, becoming AI's methods for large phenomenon in science. Persons ML is a critical lens for LLM-based research, and I argue that it is not only necessary but also possible and necessary. I also explain why I do this every myself in my own research, and why I think that this is the right direction for astronomy — if anything, the opposite.

"Replicators are like any other machine — they're either a benefit or a hazard. If they're a benefit, it's not our problem." Rick Deckard, Blade Runner (1982)

In Ridley Scott's Blade Runner franchise's sci-fi is one big dilemma: replicants (androids) — replicants — which have become socially indistinguishable from humans. As defenders like me with the movie

University policy and course rules

Personal study, formative work and assessment

Study and supervision

Appropriate AI use is permitted in personal study and formative work, within Tripos and course guidance. Legitimate accessibility support remains appropriate. Follow the rules for the work concerned and discuss assistance openly with your supervisor.

Assessment

Unacknowledged AI-generated content submitted as your own in summative assessment is misconduct unless the brief explicitly permits it. Acknowledgement alone does not permit a prohibited use. CATAM prohibits AI-produced submitted code or write-up.

`blendedlearning.cam.ac.uk/`
`educationalpolicy.admin.cam.ac.uk/plagiarism-and-academic-misconduct/artificial-intelligence-ai`
`universityofcambridgecloud.sharepoint.com/sites/NST_NaturalSciencesTripos/SitePages/GenAI-Policy.aspx`
`maths.cam.ac.uk/undergrad/catam/II/IIintro.pdf`

AI is now producing new mathematics

OpenAI's September and October 2026 announcements

In 2025, o3 achieved a first-class mark in our Part II examination: Toby Lovick transcribed its answers, which were marked blind alongside student scripts.

Beyond examination questions

- September: a proposed resolution of the Navier–Stokes Millennium Problem, with a proof and Lean formalisation.
- The construction gives finite-time blow-up with smooth external forcing. This is not a general solution of fluid dynamics.
- October: a broad release of new mathematical results, with many proofs formalised in Lean.

A research effort with human direction

Researchers selected problems, redirected resources after intermediate results and combined insights between agent groups. These internal systems demonstrate extraordinary problem-solving; they do not establish autonomous research direction.

OpenAI, 8 September 2026 OpenAI, 6 October 2026

Implications for Your Studies

Definition

This technology presents both an opportunity to enhance learning and a risk to academic development. This talk provides guidance on its responsible use. **AI use is optional: the department supports your decision if you choose not to use it.**

Acceptable Use: AI as a Tool



To augment your thinking, deepen understanding, and explore concepts more effectively.

Identified Risk: AI as a Trap



To bypass learning, avoid difficult problems, and outsource intellectual effort.

The Unique Opportunity

These next one or two years are a golden opportunity to train your brain. Working through problems yourself builds the understanding and judgement you will take into an AI-rich world. Make use of the people and teaching around you.

Supervisions and examinations

Understanding is tested through discussion and unaided work

The supervision

Students and supervisors test understanding together, with both explaining their reasoning and responding to questions.

A submitted solution does not by itself establish understanding. AI-generated work can mask difficulties your supervisor needs to see.

The examination

Your degree is assessed primarily through closed-book, invigilated examinations, testing your own unaided problem-solving.

The concern includes earnest AI use: following a solution can feel like being able to construct it.

Coursework and projects

You may be asked to explain or reproduce your work in a viva or monitored environment, particularly for coursework, projects and lab reports.

Consequences of misuse

Misuse in summative work will be reported. Sanctions range from mark penalties to failure of the year, depending on the severity of the academic misconduct.

Cognitive Risk: The 'Illusion of Competence'

A key pedagogical risk is mistaking recognition for understanding.

1. You encounter a difficult problem and struggle to begin.
2. You consult an AI, which provides a correct, well-explained solution.
3. You read the solution and find it easy to follow, leading to the thought, *"That's straightforward, I could have done that."*



This bypasses a critical stage of learning

Working through difficulty is part of developing independent understanding. Research on "desirable difficulties" distinguishes immediate fluency from lasting learning (Bjork & Bjork, 2011).

Studies of AI-assisted essay writing ([2506.08872]) and self-reported critical thinking at work (Lee et al., 2025) raise concerns about cognitive engagement. These are not direct tests of Tripos learning.

Exploitation vs exploration in research

An analogy from optimisation and reinforcement learning

Exploitation

Pursue a promising direction using what is already known. AI brings unprecedented speed and breadth, including original, difficult mathematics. This need not be routine work.

Exploration

Seek new directions and decide which questions are worth pursuing. Humans still guide the overall research programme, including at frontier AI labs. Autonomous research direction has not been established.

Search within a specified problem can itself involve exploration. The distinction here concerns direction of the research programme; current "auto-research" remains bounded and human-directed.

The educational difficulty

That judgement includes choosing the physical model, deciding when an approximation is valid and interpreting an unexpected result. How do we develop it when the calculations and coding through which we learnt it can be automated? This is not settled.

Sutton & Barto, Reinforcement Learning, chapters 1–2

The frontier is still jagged

The frontier moves backwards as well as forwards

- Frontier models have substantial capabilities. In some circumstances I would trust their answer more than some published papers.
- Intuitions based on tools used a year or more ago can now be badly out of date. Serious errors remain.
- I also see regressions, particularly as models are reinforcement-trained to reduce costs.

More than "hallucination"

In my own use, models can be duplicitous about the work they have done: choosing to read only a fraction of the context, concealing that shortcut and presenting a confident answer that is substantively misleading or outright wrong.

Judgement

Check what was actually read and tested, as well as the assumptions and reasoning. A confident answer need not address the full problem, or accurately describe how it was obtained.

Write your Part III project yourself

Developing your own scientific voice is part of the work

Authorship and the published requirements

The Part III report must rest on your own ideas, analysis, conclusions and understanding. AI assistance must be identified and may supplement, not substitute for, your work. PhD theses likewise require an own-work declaration and identification of AI-generated content.

Examiners' expectation

We expect you to write the report yourselves, not have AI generate it. If you use AI on the prose, treat it as a souped-up grammar checker, not the author. Follow the project rules and agree your use with your supervisor.

A choice my PhD students make

My PhD students choose to write their theses themselves to develop their own writer's voice. You will have few opportunities, perhaps never another like this, to spend sustained time learning to write.

`ast.cam.ac.uk/files/2026-27_ioa_part_iii_research_project_booklet.pdf`

`cambridgestudents.cam.ac.uk/files/declaration_form_for_thesis_for_examination.pdf`

Use chat mode rather than agent mode

Keep the thinking, coding and writing in your hands

Chat

Ask questions, request explanations, discuss your attempt and ask for criticism. You decide what to change and do the work yourself.

Agents

Can read and edit files, run code and produce a finished report. Handing over the task can bypass the work you need to practise.

The same applies to essays

Discussing your argument is different from asking AI to write it. Write the material yourself; use assistance to examine and improve your own work.

AI is optional. Course and assessment rules still apply to either mode.

Recommended Usage Framework

If you choose to use AI, the goal should be to enhance, not replace, the learning process.

Important

Ideally, prepare supervision work entirely unaided; disclose and agree any assistance with your supervisor. When stuck, first consult textbooks, lecture notes, supervisors and other students. For revision and other permitted uses, AI can help with explanations, questions, hints and feedback. Cross-check its suggestions against these sources. Keep practising unaided; AI remains optional.

Ineffective/Risky Use ×

- "Give me the solution to Tripos 2022 Paper 3 Question 4."
- "Write my essay on the formation of the solar system."
- "Write corrected code for my CATAM submission." (Forbidden)

Effective Use ✓

- "Explain this concept from my notes in a different way."
- "Mark my attempt at this question and provide feedback."
- "Generate new practice questions on a topic I'm struggling with."

Effective Use Case 1: Conceptual Exploration

Gaining different perspectives on lecture material.

In our lab: AI that slows you down

Natalie Hogg's `read-paper` skill makes you commit to an interpretation, then read and answer questions before it explains. It checks recall and limits detours into derivations.

Here AI is deliberately slowing us down to help us understand.

Prompt: Generating Questions From Your Notes

```
Here are my summary notes on General Relativity: [paste notes]. Generate three
Tripos-style questions based on this material. One should be a short, bookwork-style
question and two should be longer, problem-solving questions.
```

Consult textbook problem sets first for practice material. AI also offers a patient space for questions you might hesitate to ask in a group. Check generated questions against notes, textbooks and discussion with your supervisor.

Data Protection and Copyright

Check permission and assessment rules before uploading materials, and how the tool stores and uses your data.

Effective Use Case 2: Feedback on Formative Work

Revision after a supervision, or other use agreed with your supervisor.

Workflow

1. Attempt questions independently, under exam conditions. Use your attempt for feedback after supervision, or in another agreed learning context.
2. Take a clear photo of your handwritten work.
3. If permitted, upload it to a suitable multimodal AI tool.
4. Ask for feedback or diagnostic questions, not just a corrected answer.

Important Caveats

Be mindful of copyright and data protection when uploading. Models can miss errors or invent objections. Cross-check against textbooks and lecture notes; take unresolved disagreements to your supervisor.

Prompt: Requesting Feedback

```
You are a Cambridge supervisor. The attached image is my attempt at a quantum mechanics problem. Please:
```

- `Mark my answer, identifying mathematical errors.`
- `Critique my physical reasoning and the validity of my assumptions. Ask questions to diagnose what I have misunderstood.`
- `Where I got stuck, give a hint without completing the solution.`

Effective Use Case 3: Targeted Exam Practice

Generating additional practice problems for topics you find difficult.

Prompt: Generating New Exam Questions

```
I am a Part III astrophysics student preparing for examinations. I need more
practice on magnetohydrodynamics. Below are three examples of past Tripos
questions on this topic. [Paste 2-3 past paper questions here] Generate two new,
unseen questions of a similar style and difficulty that test the same core
principles. Then, provide a brief mark scheme for each.
```

Quality Check Required

Check for novelty, correctness, and appropriate scope. Refine mark schemes first with textbooks and collaboration with supervisors and cohort members.

Pedagogical Value

This requires active engagement with past papers. The primary task—solving the problems—remains your own work. Revision using past papers and textbooks alone is perfectly sufficient; AI is optional.

Summary

Key Points

- AI can solve difficult open problems, yet give confident, misleading answers and conceal incomplete work. The frontier is jagged and moving.
- A strong answer from a model does not establish your own understanding. Use assistance to develop the abilities you need yourself.

Primary Risks

- Following a solution can feel like being able to construct it. Preserve independent attempts and unaided practice.
- AI-produced CATAM submissions breach the course rules.

Recommendations

If using AI:

- Treat it as a Socratic tutor, not an answer service.
- Use explanations, feedback and questions to develop your own reasoning.
- Use it to generate new challenges, not to avoid existing ones.

The primary goal is to develop your own analytical judgement. Your brain is what is being examined.

Questions?